All you need to know about relational hyper-event modeling

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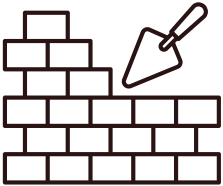
Winter School in
Network Data Science and Artificial Intelligence
February 10, 2026

Università
della
Svizzera
italiana

Section	Presenter	Time	Abstract
Core RHEMs	Melania	14:00 – 14:30	computing hyperevent covariates; formulating, estimating, and interpreting linear RHEMs.
Practical (1)	Martina	14:35 – 15:15	computing RHEM statistics using `eventnet`.
Non-Linear RHEMs	Martina	15:20 – 15:50	modeling non-linear, time-varying covariate effects while accounting for network and actor heterogeneity.
Practical (2)	Martina	15:55 – 16:45	applying RHEMs to empirical datasets using `R Studio`.

What did we learn yesterday?

1. **Core REMs**: goal, core formulation, and key definitions of covariates.
2. **REM inference**: how to estimate and interpret the effects of explanatory variables on event dynamics.
3. **Non-linear REMs**: exploiting more complex model specifications, including approaches that can reduce computational costs.
4. **Practical**: flexibility of REMs in explaining real-world phenomena such as alien species invasions and bike-sharing systems; fitting REMs in R.



... so, what is the plan for today?

1. **Core RHEMs**: extending most notions to *relational hyper event modelling* (RHEM)... after Jürgen's talk, this should already feel familiar! ☺
2. **Non-linear REMs**: increasing (even more!) the *complexity* of the model formulation.
3. **Practical**: flexibility of RHEMs in explaining empirical phenomena such as *co-appearance in an artwork* and *scientific innovation*; learning how to use `eventnet` and fitting RHEMs in `R`.

Why is an extension needed?

We will refer to two main guiding examples:

1. ***Les Misérables***: *co-appearance* of actors in one or more chapters ⇒ each chapter is a *hyperevent*.
2. **Scientific innovation**: *publication* of an article by a group of authors citing a set of articles ⇒ each new paper is a *hyperevent*.

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   Who are the senders, and who are the receivers?

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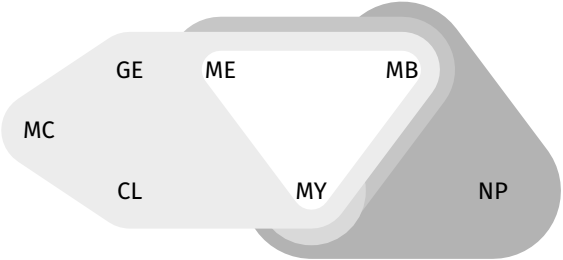
1. How do these two types of hyperevents differ?
   Who are the senders, and who are the receivers?
2. Can you think of an example of a hyperevent with a single sender and multiple receivers?
3. How were these events treated before the introduction of RHEMs?
   Do you have suggestions for converting them into dyadic relational events?

Let us look at each example: 1. *Les Misérables*

Les Misérables: each chapter is an hyperevent.
co-appearance of actors in one or more chapters

1	event.id	participant	type
110	01.01.01	MY	chapter
111	01.01.01	NP	chapter
112	01.01.01	MB	chapter
113	01.01.02	MY	chapter
114	01.01.02	ME	chapter
115	01.01.02	MB	chapter
116	01.01.03	MY	chapter
117	01.01.04	MY	chapter
118	01.01.04	ME	chapter
119	01.01.04	CL	chapter
120	01.01.04	GE	chapter
121	01.01.04	MC	chapter
122	01.01.04	MB	chapter
123	01.01.05	MY	chapter
124	01.01.05	MB	chapter
125	01.01.05	ME	chapter



? : Discussion time!

What would be the goal of a RHEM here?

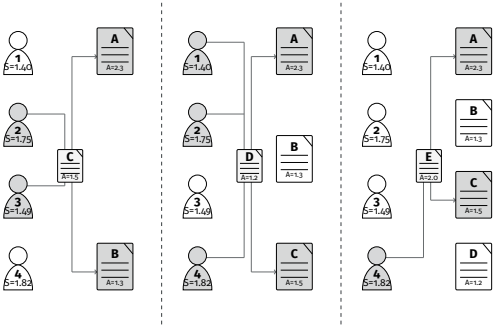
$$E = \{(\{\text{MY, NP, MB}\}, t_1), (\{\text{MY, ME, MB}\}, t_2), (\{\text{MY}\}, t_3), (\{\text{MY, ME, CL, GE, MC, MB}\}, t_4), (\{\text{MY, ME, MB}\}, t_5), \dots\}$$

Let us look at each example: 2. Scientific innovation

Scientific innovation: each paper publication is an hyperevent.
a group of authors publishes a paper citing a set of papers

Synthetic data as an example

Time	Authors	Paper	Citations
1950	$\{2,3\}$	C	$\{A,B\}$
1951	$\{1,2,4\}$	D	$\{A,C\}$
1953	$\{4\}$	E	$\{A,C\}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$



? : Discussion time!

Examples of exogenous covariates, here?

$$E = \{(\{2,3\},\{A,B\},t_1 = 1950),(\{1,2,4\},\{A,C\},t_2 = 1951),(\{4\},\{A,C\},t_3 = 1953)\}$$

Relational Hyper Event Modelling (RHEM)

Let $\{N_{SR}(t)\}_{S \subseteq V^S, R \subseteq V^R, t \geq 0}$ be a **counting process** such that:

- $N_{SR}(t)$ = number of events $S \to R$ until time t.
- $N_{SR}(0) = 0$

According to **Doob-Meyer theorem**,

$$N_{SR}(t) = \underbrace{\Lambda_{SR}(t)}_{\text{predictable part}} + \underbrace{M_{SR}(t)}_{\text{noisy part}} = \int_0^t \lambda_{SR}(u)\,\mathrm{d}u + M_{SR}(t)$$

where $\lambda_{SR}(t)$ is the **instantaneous rate** of $S \to R$ occurring at time t.

$$\lambda_{SR}(t) = \underbrace{W_{SR}(t)}_{\text{risk indicator}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \exp\Big\{ \underbrace{f\left(\boldsymbol{x}_{SR}(t), t\right)}_{\text{contribution function}} \Big\}$$

Linear Model: $f\left(\boldsymbol{x}_{SR}(t), t\right) = \boldsymbol{\beta}^\top \boldsymbol{x}_{SR}(t)$ – $\boldsymbol{\beta}$ fixed in time and covariate value

Let us focus on $x_{SR}(t)$... from attributes to covariates

$\boldsymbol{x}_{SR}(t)$ should capture information about the instantaneous rate of (S, R, t).

1. Iterate over specific subsets $S' \subseteq S$.
2. Compute a numerical property $z_{S'}(t)$.
3. Aggregate the values of $z_{S'}(t)$ over the subsets S'.

What does this mean in practice? Let us see two examples!

Exogenous information can be used for several goals...

Node-level feature: $z_s(t)$ is a node attribute, e.g. gender - binary (at the time of Les Misérables...).

Hyperedge covariate: avg. of z_s over nodes in S; $x_S(t) = \text{avg.z}(S,t) = \sum_{s \in S} z_s(t)/|S|$

Feature (dis)similarity: diff.z measures diversity; a negative parameter $\Rightarrow$ **homophily**.

$$\text{diff.z}(S,t) = \sum_{\{s,s'\} \in \binom{S}{2}} \frac{|z_s(t) - z_{s'}(t)|}{\binom{|S|}{2}}$$

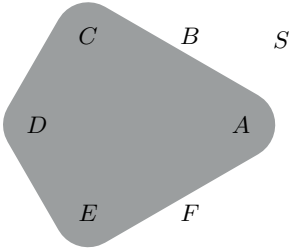
$$\text{avg.z}(S_1) = 0 \qquad \text{avg.z}(S_2) = 0.5 \qquad \text{avg.z}(S_3) = 1$$
$$\text{diff.z}(S_1) = 0 \qquad \text{diff.z}(S_2) = \tfrac{9}{15} \qquad \text{diff.z}(S_3) = 0$$

Endogenous $x_{SR}(t)$... Subset-repetition

Subset repetition of order ρ: average # of prior joint events over $S' \subseteq S$ of size ρ.

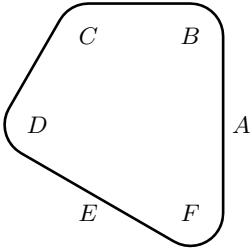
$$\text{hy.deg}(S', t) = \sum_{e_i : t_i < t} \mathbf{1}(S' \subseteq S_i)$$

$$\text{sub.rep}^{(\rho)}(S, t) = \sum_{S' \in \binom{S}{\rho}} \frac{\text{hy.deg}(S', t)}{\binom{|S|}{\rho}}$$



$S = \{B, C, D, F\}$

$$\text{sub.rep}^{(1)}(S) = 2/4$$
$$\text{sub.rep}^{(2)}(S) = 1/6$$
$$\text{sub.rep}^{(3)}(S) = 0/4$$



Let's move to $f(\boldsymbol{x}_{SR}(t), t)$... the contribution function

Do you remember?

$$\lambda_{SR}(t) = \underbrace{W_{SR}(t)}_{\text{risk indicator}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \exp\Big\{ \underbrace{f\left(\boldsymbol{x}_{SR}(t), t\right)}_{\text{contribution function}} \Big\}$$

- We explored **covariates** $\boldsymbol{x}_{SR}(t)$: **structures** based on **past events** or external information.
- We now focus on their **effect**: their influence on the log-event rate.

The simplest form of effect is a **fixed linear effect** (FLE):

$$f^{\text{FLE}}\big(x_{SR}(t), t\big) = \beta \cdot x_{SR}(t)$$

We have already seen that we need to go *beyond linearity*.

What is eventnet?

https://github.com/juergenlerner/eventnet

Computes specified covariates for events and non-events.

observed events

e_1	A	t_1
e_1	B	t_1
e_1	C	t_1
e_2	D	t_2
e_2	E	t_2
...		

→ **eventnet** →

statistics of events and non-events

e_1	$A\|B\|C$	t_1	1	2.4	0.7
e_1	$A\|B\|D$	t_1	0	0.1	1.5
e_1	$A\|B\|E$	t_1	0	0.5	1.9
e_1	$C\|D\|E$	t_1	0	0.3	0.9
e_2	$D\|E$	t_2	1	1.9	0.1
e_2	$A\|B$	t_2	0	2.1	1.9
e_2	$C\|E$	t_2	0	0.7.1	0.2
...					

configuration: determines how statistics are computed from input events (reproducibility)

Configurations can be specified with the eventnet graphical user interface or via a configuration file (XML).

eventnet workflow – more details

past events

A, B, C meet at t_1

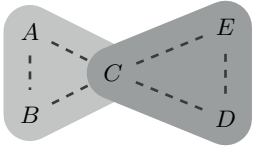
C, D, E meet at t_2

...

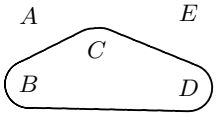
all events

before t

hyperedge attributes



hyperedge statistics



partial rep, closure, ...

specify

prob of next event(s)

do B, C, D meet at t?

param. function of statistics

Hands-on: Part 1

$\approx$ 45 minutes

- Format of the input data.
- CSV and time settings.
- Specifying network attributes and statistics.
- Specifying the observed events, risk set, and sampling.
- Perform case-control partial likelihood inference.
- Fit a linear RHEM in R.

Directed hyperevents

Research settings about

- **teams of actors** collaboratively doing **work**
- building on **prior work** (of the same or different actors).

Examples include:

- Scientists publishing papers that cite previous papers.
- Inventors filling patents referencing other patents.
- Artists producing music / movies making implicit or explicit references to other music / movies.

Case study

Data: https://www.aminer.org/citation.

DBLP-Citation-network V14, restricted to journal papers.

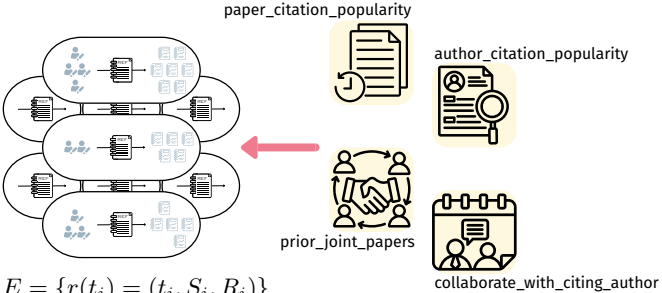
1,416,353 papers written by 1,286,941 unique authors

References:

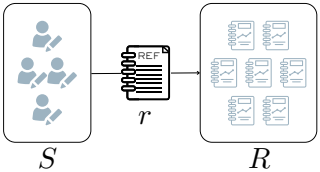
1. Lerner, Hâncean, & Lomi: **Relational hyperevent models for the coevolution of coauthoring and citation networks**.
   [Lerner et al., 2025]
2. Boschi, Lerner, & Wit: **Beyond linearity and time-homogeneity: relational hyper event models with time-varying non-linear effects**.
   [Boschi et al., 2025]

Drivers of Scientific Innovation

Relational Hyperevent: publication in year t of article $r(t)$ by a group of authors S citing articles R



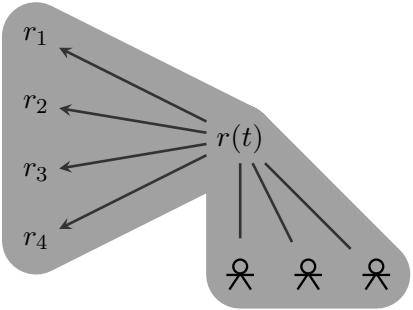
$E = \{r(t_i) = (t_i, S_i, R_i)\}$
$i = 1, \ldots, 1\,416\,353$



Relational Hyper Event Model: captures how key covariates influence the rate of hyperevent occurrences.

The unit of analysis.

Work $r(t)$ produced at time t by **team** of actors $S = \{s_1, s_2, s_3, \dots\}$ referencing a **list** of previous works $R = \{r_1, r_2, r_3, r_4, \dots\}$.



Previous works produced by same, overlapping or disjoint teams; referencing other previous works, produced by ...

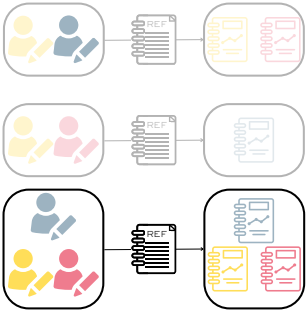
Subset repetition of order (ρ, ℓ)

Covariates $\boldsymbol{x}_{SR}(t)$: statistics derived from **past events** and/or **exogenous information**. How do these covariates change in the case of **directed hyperevents**?

⚙: E.g. **Subset repetition**:

$$\text{cite}^{\text{aut-pap}}(S, R, t) = \sum_{t_i < t} \omega(t - t_i) \cdot 1_{\{S \subseteq S_i \cap R \subseteq R_i\}}$$

$$\text{subrep}^{(\rho,\ell)}(S, R, t) = \sum_{(S',R') \in \binom{S}{\rho} \times \binom{R}{\ell}} \frac{\text{cite}^{\text{aut-pap}}(S', R', t)}{\binom{|S|}{\rho} \times \binom{|R|}{\ell}}$$



Types of effects / research questions.

Effects in the collaboration network.

- Activity of authors; homophily effects.
- Repeated collaboration among pairs, triples, ...of authors.
- Closure: collaborating with collaborators' collaborators.

Effects in the reference network.

- Popularity of papers.
- Repeated co-references to pairs, triples, ...of papers.

Effects in the mixed two-mode network.

- Referencing the own paper (self-citation).
- Referencing the papers/references of collaborators.
- Collaborating with authors referencing own/same papers.
- Reference repetition / reciprocation on the author level.

E.g. **cite paper and its references** (whiteboard).

Before the practice...

Do you remember?

$$\lambda_{SR}(t) = \underbrace{W_{SR}(t)}_{\text{risk indicator}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \exp\Big\{ \underbrace{f\left(\boldsymbol{x}_{SR}(t), t\right)}_{\text{contribution function}} \Big\}$$

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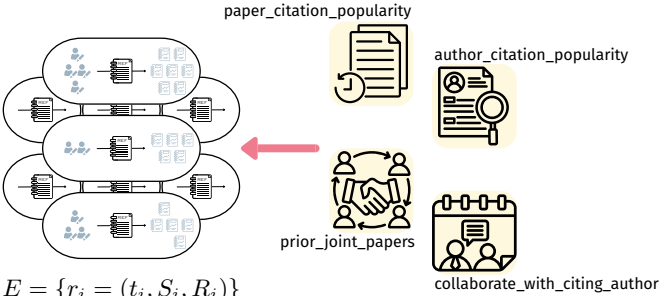
We have already seen that we need to go **beyond linearity**.

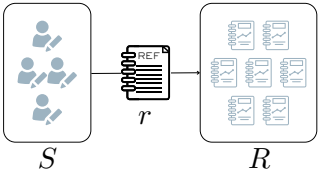
	parameters
num prior papers	$0.129\ (0.003)^{***}$
diff prior papers	$0.185\ (0.002)^{***}$
num prior joint papers	$0.823\ (0.004)^{***}$
author citation popularity	$-0.017\ (0.001)^{***}$
diff auth cite pop	$-0.007\ (0.001)^{***}$
collab w citing author	$0.318\ (0.003)^{***}$
closure: common coauthor	$-0.014\ (0.001)^{***}$
closure: citing common paper	$-0.112\ (0.002)^{***}$
closure: citing common author	$0.001\ (0.001)$
closure: cited by common author	$0.007\ (0.001)^{***}$
paper citation popularity	$0.170\ (0.000)^{***}$
paper-pair cocitation	$0.032\ (0.001)^{***}$
paper-triple cocitation	$-0.170\ (0.004)^{***}$
author citation repetition	$0.643\ (0.008)^{***}$
paper outdegree popularity	$0.195\ (0.001)^{***}$
cite paper and its refs	$7.360\ (0.010)^{***}$
adopt citation of coauthor	$-0.264\ (0.003)^{***}$
self citation	$1.799\ (0.012)^{***}$
cite coauthor's paper	$0.291\ (0.007)^{***}$
author-author citation repetition	$-0.381\ (0.007)^{***}$
author-author citation reciprocation	$-0.343\ (0.007)^{***}$

Are drivers of Scientific Collaboration truly Constant?

Relational Hyperevent: publication in year t of article $r(t)$ by a group of authors S citing articles R



$E = \{r_i = (t_i, S_i, R_i)\}$
$i = 1, \ldots, 1\,416\,353$



Relational Hyper Event Model: captures how key covariates influence the rate of hyperevent occurrences.

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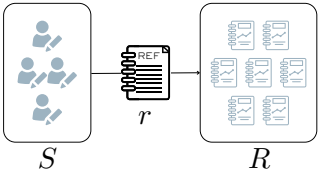
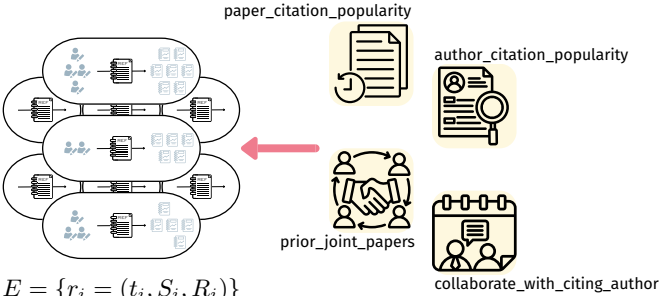


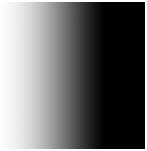


$E = \{r_i = (t_i, S_i, R_i)\}$
$i = 1, \ldots, 1\,416\,353$

Relational Hyper Event Model: captures how key covariates influence the rate of hyperevent occurrences.

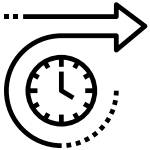
? : Discussion time! Any idea why this should not be the case?

What can break Linearity and Time-Homogeneity?



Saturation: individuals face cognitive overload and become **unable to effectively additional information**.
💡: *allow drivers' influence to exhibit a non-linear pattern.*

Forgetfulness: propensity to **lose track of past details** over time.
💡: *define mechanisms as interval times with non-linear effect.*



External Changes: **shifts** in the environment, technology, or social norms that **alter how factors influence system**.
💡: *allow drivers' influence to have a time-varying effect.*

TVE and NLE for RHEMs

Contribution function:

$$f\left(x_{SR}(t), t\right)$$

- **Effect**: influence of the covariate on the log-rate of event occurrence.
  - **Fixed linear effect** (FLE):

$$f^{\text{FLE}}\big(x_{SR}(t), t\big) = \beta \cdot x_{SR}(t)$$

  - **Non-linear effect** (NLE): captures *saturation* and *forgetfulness*.

$$f^{\text{NLE}}\big(x_{SR}(t)\big) = \sum_{q=1}^{Q} \beta_q \underbrace{b_q\big(x_{SR}(t)\big)}_{\text{basis function evaluation of covariate}}$$

  - **Time-varying effect (TVE)**: an *effect may change over calendar time* due to *changing circumstances* or *unmeasured confounders*.

$$f^{\text{TVE}}\big(x_{SR}(t), t\big) = \alpha(t) \cdot x_{SR}(t), \qquad \alpha(t) = \sum_{l=1}^{L} \alpha_l \underbrace{a_l(t)}_{\text{basis function evaluation of time}}$$

Joint time-varying non-linear effect...

Proposal: employ **tensor product smooth** (Wood, 2016) to model TVNLE in RHEM.

$$f^{\text{TVNLE}}(x_{SR}(t), t) = \sum_{q=1}^{Q} \left(\underbrace{\sum_{l=1}^{L} \alpha_{ql} a_l(t)}_{\text{time-varying coefficient}} \right) \underbrace{b_q(x_{SR}(t))}_{\text{basis function evaluation of covariate}}$$

☺: coefficients of basis evaluations of covariates are not fixed but vary over time.

Joint time-varying non-linear effect...

Proposal: employ **tensor product smooth** (Wood, 2016) to model TVNLE in RHEM.

$$f^{\text{TVNLE}}(x_{SR}(t), t) = \sum_{q=1}^{Q} \left(\underbrace{\sum_{l=1}^{L} \alpha_{ql} a_l(t)}_{\text{time-varying coefficient}} \right) \underbrace{b_q(x_{SR}(t))}_{\text{basis function evaluation of covariate}}$$

☺: coefficients of basis evaluations of covariates are not fixed but vary over time.

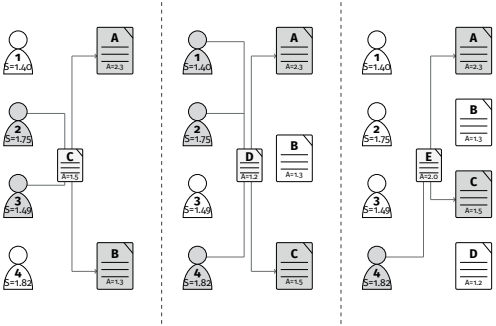
? : **Discussion time!** How would you represent this type of effect?

Joint time-varying non-linear effect... in practice I

Scientific innovation: each paper publication is an hyperevent.
a group of authors publishes a paper citing a set of papers

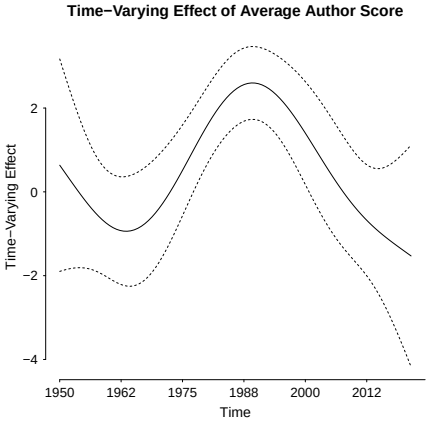
Synthetic data as an example

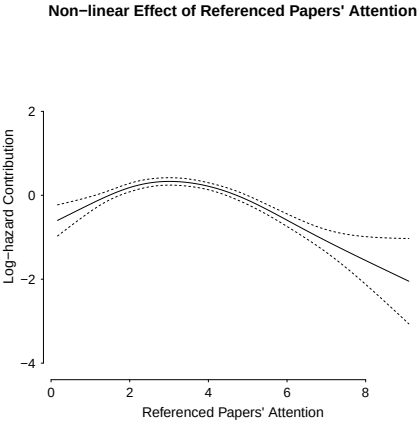
Time	Authors	Paper	Citations
1950	$\{2,3\}$	C	$\{A,B\}$
1951	$\{1,2,4\}$	D	$\{A,C\}$
1953	$\{4\}$	E	$\{A,C\}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$



$$E = \{(\{2,3\},\{A,B\},t_1 = 1950),(\{1,2,4\},\{A,C\},t_2 = 1951),(\{4\},\{A,C\},t_3 = 1953)\}$$

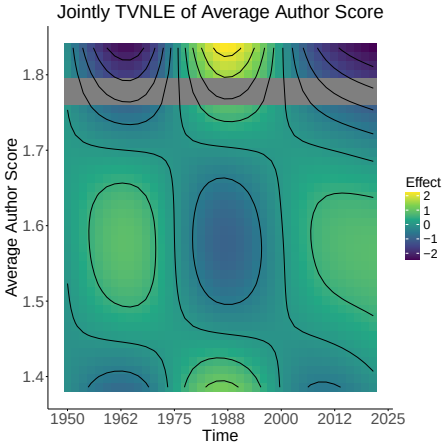
Let's start with TVE and NLE...

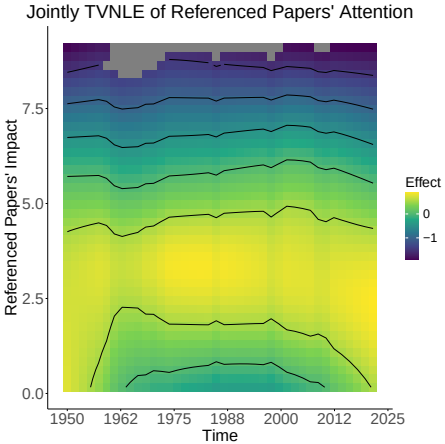




? : **Discussion time!** Do you remember how to fit these effects in R Studio? How do we interpret them?

Let's now fit a TVNLE...





? : Discussion time! Do we need a TVNLE effect in both cases?

Step 1: compute covariates with `eventnet`

Using the `eventnet` software:

1. Extract event information from the raw data.
2. Specify the components required by `eventnet` to define the risk set.
3. Sample a non-event observation from the risk set.
4. Compute covariates for both events and non-events.

Case-control dataset: Merge the output of `eventnet` by event and non-event identifiers so that each row contains information for both event and non-event.

Step 2: compute covariates with `eventnet`

1. Apply a monotonic transformation to each covariate x:

$$\dot{x} = 1 - \exp\left(-\frac{x}{\epsilon}\right)$$

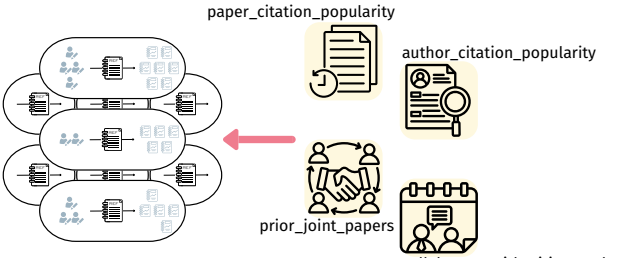
   - Choose ϵ so that the transformed covariate (evaluated for events) is as close as possible to a uniform distribution.
   - Apply the same transformation to both events and non-events.

2. Transform event times t using the empirical cumulative distribution function:

$$\dot{t} = \mathrm{ecdf}_t(t)$$

Coauthorship Citation Networks

RHEM extension applied to the DBLP-Citation-network V14 dataset (journal articles).



$$E = \{r_i = (t_i, S_i, R_i)\}$$
$$i = 1, \ldots, 1\,416\,353$$

1. *Prior papers* ;
2. *Prior joint papers*;
3. *Paper citation popularity* ;
4. *Paper pair co-citation*;
5. *Author citation repetition*;
6. *Cite paper and its references*;
7. *Difference in prior papers* ;
8. *Author citation popularity*;
9. *Difference in author citation popularity*;
10. *Collaborate with citing author*;
11. *Author self citation* ;
12. *Paper outdegree popularity* ;

Author Self Citation - Covariate

- STEP 1: **Define the k-th covariate.**

$$\text{author}(s,r,t) = \sum_{t_i<t} \omega(t-t_m) \cdot \mathbb{1}_{\{s=j_i \,\wedge\, r\in S_i\}}$$

$$x^{(k)} = \text{author_self_citation}(S,R,t) = \sum_{s\in S,\, r\in R} \frac{\text{author}(s,r,t)}{|S|\cdot|R|}$$

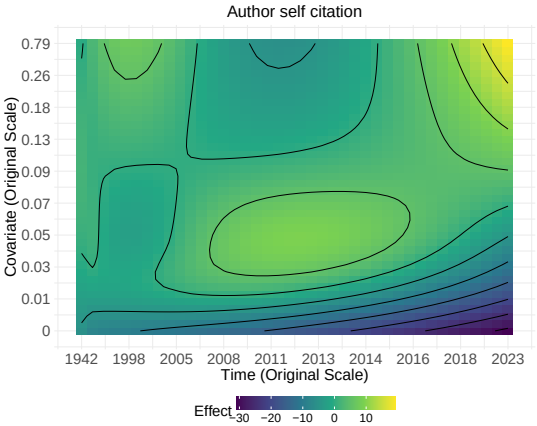
- STEP 2:
  - **Transform the k-th covariate**, to *improve model fitting stability.*

$$\dot{x}^{(k)} = 1 - \exp\left(-\frac{x^{(k)}}{\epsilon^{(k)}}\right),$$

    where $\epsilon^{(k)}$ is selected through an optimisation procedure.
  - **Transform time**, to *improve model fitting stability.*

$$\dot{t} = \text{ecdf}_t(t)$$

Author Self Citation - Effect



- monotone effect at the beginning and end of the observation period, only.
- **between 2008 and 2016, the effect is non-monotonic**:
  it first increases with the covariate value, then decreases.

Scientific innovation

Hands-on: part 2

$\approx$ 45 minutes

- Load the data in `R Studio`.
- Create the case-control dataset.
- Fit models with NLE, TVE, and TVNLE.

Today was quite an intense RHEM day...

- Tensor product smooths can be employed to model TVNLE dynamics in RHEMs.
- Joint TVNLE models are a powerful tool for uncovering not only non-linear but also non-monotonic patterns that **linear models would fail to capture**.
- Applied this approach to study the **dynamics of citation and impact**.
- **Author self-citation**:
  - Citing a certain share of one's own work seems natural and nearly unavoidable;
  - Excessive self-citation could be perceived negatively by the research community.

Relational events that have an outcome.

Sometimes, relational (hyper)events produce a tangible outcome with associated "success" or "performance".

- Teams of scientists publishing an article → impact.
- Filmmakers producing a film → revenue, popularity, evaluation, awards.
- Medical teams performing an operation → cost, duration, benefits for the patient.
- Two soccer teams meeting in a match.

Past success may explain future collaboration (RHEM).

Past success may explain future success (RHOM).

Lerner & Hâncean (2023). **Micro-level network dynamics of scientific collaboration and impact: relational hyperevent models for the analysis of coauthor networks.** *Network Science*.
[Lerner and Hâncean, 2023]

Data: relational events that have an outcome.

A relational hyperevent with an outcome is given by

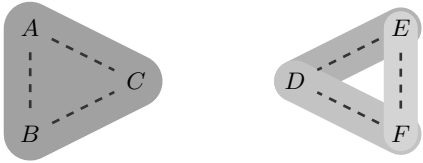
$$e_i = (S_i, t_i, x_i, y_i) \ ,$$

- t_i is the time of the event;
- $S_i \subseteq V_{1,t_i}$ event participants (e. g., coauthors);
- x_i covariates of the event (type, weight);
- y_i is the *relational outcome* (success, performance, impact).

Recall: subset repetition of order k.

Average number of prior joint events over $S' \subseteq S$ of size S.

$$\text{sub.rep}^{(k)}(S,t) \quad = \quad \sum_{S' \in \binom{S}{k}} \frac{\text{hy.deg}(S',t)}{\binom{|S|}{k}} \quad .$$



$\text{sub.rep}^{(1)}(\{A,B,C\}) = 1$ $\quad$ $\text{sub.rep}^{(1)}(\{D,E,F\}) = 2$

$\text{sub.rep}^{(2)}(\{A,B,C\}) = 1$ $\quad$ $\text{sub.rep}^{(2)}(\{D,E,F\}) = 1$

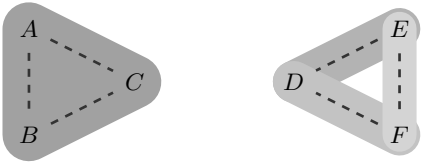
$\text{sub.rep}^{(3)}(\{A,B,C\}) = 1$ $\quad$ $\text{sub.rep}^{(3)}(\{D,E,F\}) = 0$

$$\text{hy.deg}(S',t) \quad = \quad \sum_{e_i : t_i < t} \mathbf{1}(S' \subseteq S_e)$$

Prior (shared) success of order k.

Average weight of prior joint events over $S' \subseteq S$ of size k.

$$\text{prior.success}^{(k)}(S,t) = \frac{\sum_{S' \in \binom{S}{k}} \text{performance}(S',t)}{\sum_{S' \in \binom{S}{k}} \text{hy.deg}(S',t)} \ .$$

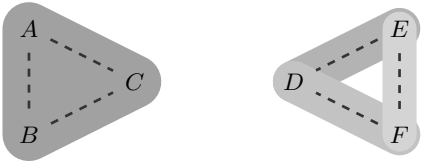


$$\text{performance}(S',t) = \sum_{e_i : t_i < t} y_i \cdot \mathbf{1}(S' \subseteq S_i) \ .$$

Related: prior success disparity.

Individual success vs. shared success.

Assume that two teams are composed of individuals that have been equally successful in the past.



It still makes a difference whether they have **jointly** achieved this success.

A team is more than the sum of its members.

Recall: relational events that have an outcome.

A relational hyperevent with an outcome is given by

$$e_i = (S_i, t_i, y_i) \ ,$$

- t_i is the time of the event;
- $S_i \subseteq V_{1,t_i}$ event participants (e. g., coauthors);
- y_i is the *relational outcome* (success, performance, impact).

RHEM explain the participants S_i (and/or the time t_i).

RHOM explain the outcome y_i, given that there is an event (S_i, t_i).

Relational hyperevent outcome models (RHOM).

Given of sequence of relational hyperevents with an outcome

$$E = [(S_1, t_1, y_1), \ldots, (S_n, t_n, y_n)]$$

RHOM specify regression models (OLS, logit, ...) for

$$y_i \sim f(t_i, S_i; \mathcal{W}_{t_i^-})$$

Allows to control for certain forms of dependence:

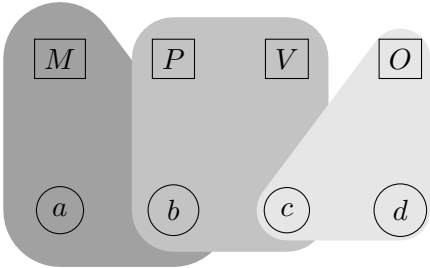
- repeated observations by the same team;
- observations with overlapping teams;

R(H)EM and R(H)OM can be specified with the same explanatory variables, allowing to compare effects on **team selection** and **team performance**.

Relational events that are associated with labels.

Actors co-participating in crime events of certain type:
market (M), property (P), violence (V), “other” (O).

$$e_1 = ([\{a, b\}, \{M\}], t_1)$$
$$e_2 = ([\{b, c\}, \{P, V]\}, t_2)$$
$$e_3 = ([\{c, d\}, \{O\}], t_3)$$

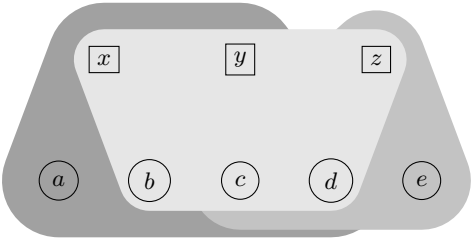


In general, meeting, communication, or team work events might have associated “labels” (purposes, objectives, topics, ...).

Two-mode relational hyperevent network.

Actors (circles: $a, b, \ldots$) participate in events having labels (squares: $x, y, \ldots$).

$$e_1 = ([\{a, b, c, d\}, \{x, y\}], t_1)$$
$$e_2 = ([\{c, d, e\}, \{z\}], t_2)$$
$$e_3 = ([\{b, c, d\}, \{x, y, z\}], t_3)$$



Many additional effects possible:
Partial repetition: of actors, labels, and mixed.
Closure: in the space of actors, labels, and mixed.

References I

Martina Boschi, Jürgen Lerner, and Ernst C Wit. Beyond linearity and time-homogeneity: relational hyper event models with time-varying non-linear effects. *arXiv preprint arXiv:2509.05289*, 2025.

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